

The Best Open Source Storage Solutions: An Overview

Millions of bytes of data are being generated every moment, every day, around the world, and this volume will only increase in the future. The storage of all these bytes has become a challenge. Simple storage is no longer enough—the data that is stored has to be safe, secure and retrievable. Of the many data storage solutions available today, this article evaluates the best open source storage solutions.



Digital storage needs are increasing rapidly. According to IDC, there will be 40 zetta bytes of data in the digital universe by 2020. The open source community has developed a number of products to assist users to store, secure and manage all that data. Open source software defined storage provides the benefits of open source tools and the benefits of massive economies of scale by running the products on commodity hardware. The end result is scalable, agile, easily manageable, and loosely coupled environments for unstructured data storage. Software defined storage brings the power of virtualisation to data storage. It allows organisations to abstract and pool storage capacity, as well as compute and networking resources, across on-premise and cloud environments.

Here is an overview of nine open source storage solutions that are considered top-of-the-line today.

Gluster: A NAS/SAN solution

Gluster is an open source and general-purpose distributed file system that aggregates storage exports over the

network to provide a single unified namespace. It is written in C, with a GPLv3 licence and access APIs - libglusterfs and FUSE. It pools storage servers over TCP/IP or InfiniBand Remote Direct Memory Access, which allows rapid provisioning of additional storage based on storage requirements. It can be used with low-cost commodity computers. Some of the popular uses of GlusterFS are in: unstructured data storage, virtual machine image storage, archiving, disaster recovery, cloud storage for service providers, semi-structured and structured data, Big Data, content delivery and streaming media. It is very suitable for unstructured data such as audio and video files, images, documents and log files. Gluster's open source community develops GlusterFS and related products.

GlusterFS is based on the following concepts:

- **Trusted Storage Pool** is a collection of storage servers that work as a logical partition for all data and management operations
- **Brick** is the mixture of a node and an export directory
- **Volume** is a mountable entity and logical collection of bricks